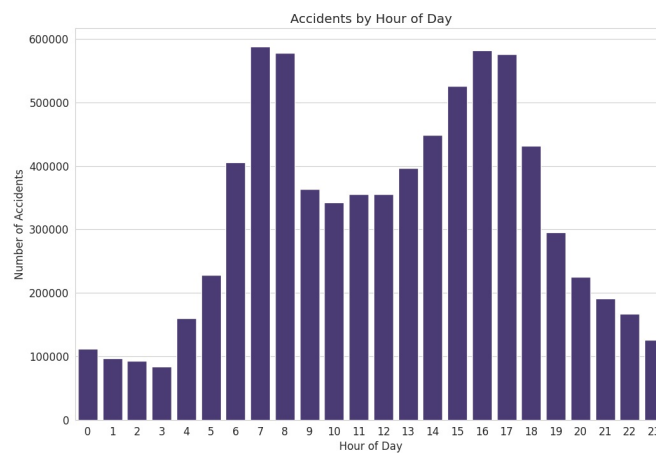


Exploratory Data Analysis Results

This document summarizes the findings from the exploratory data analysis of the US traffic accidents dataset.

1. Temporal Patterns

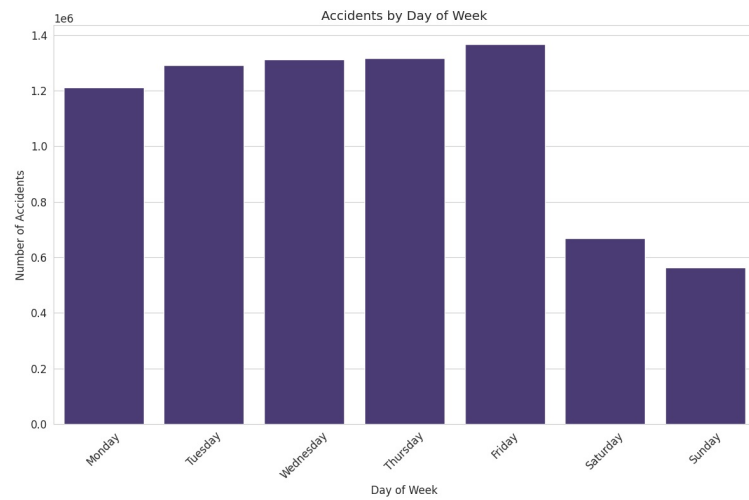
Hourly Distribution



Accidents by Hour

- Peak accident hours occur during rush hours:
 - Morning peak: 6-9 AM
 - Evening peak: 3-6 PM
- Lowest accident rates occur during early morning hours (2-4 AM)

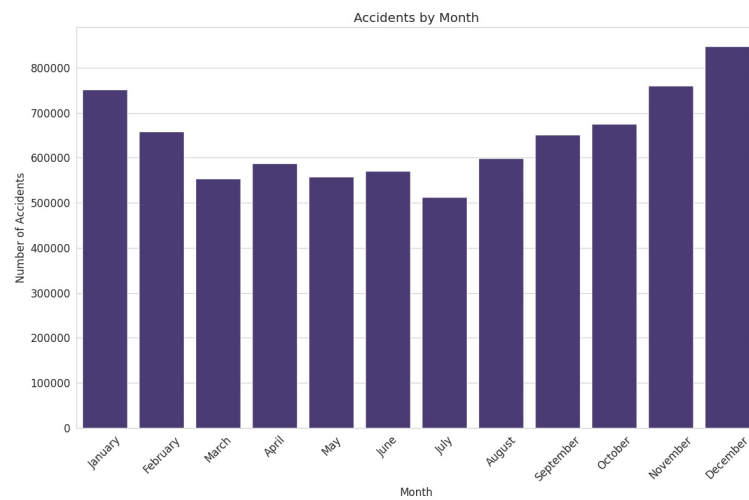
Daily Distribution



Accidents by Day

- Weekdays show higher accident rates than weekends
- Friday has the highest number of accidents
- Sunday has the lowest number of accidents

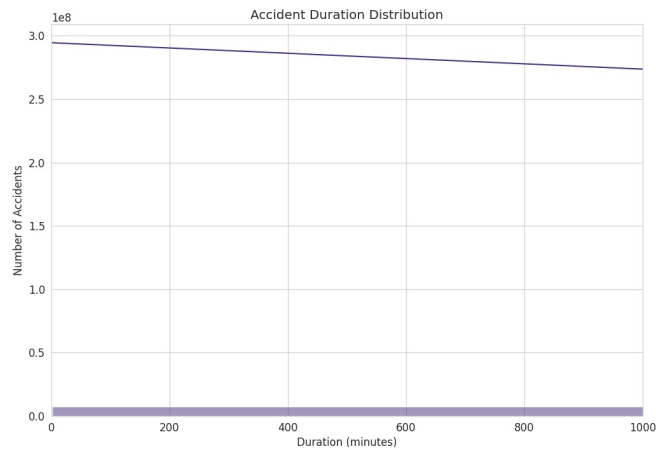
Monthly Distribution



Accidents by Month

- Higher accident rates in winter months (November-February)
- Lower accident rates in summer months (June-August)
- Gradual increase from summer to winter

Accident Duration

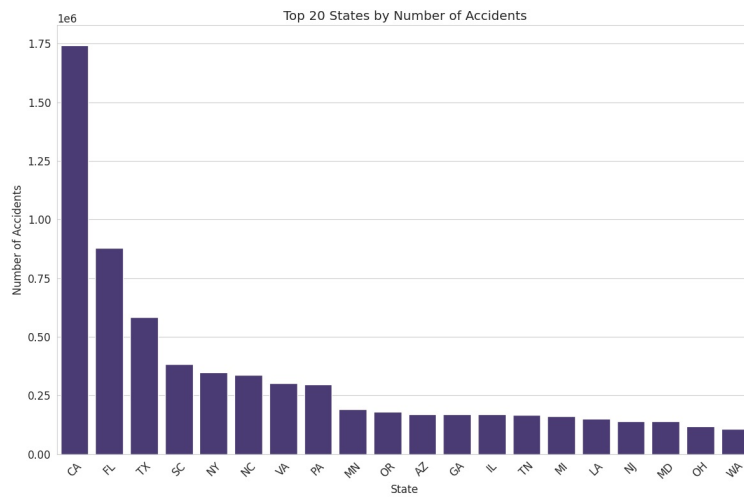


Accident Duration

- Most accidents last between 30-120 minutes
- Long-tail distribution with some accidents lasting several hours
- Median duration: ~60 minutes

2. Geographical Patterns

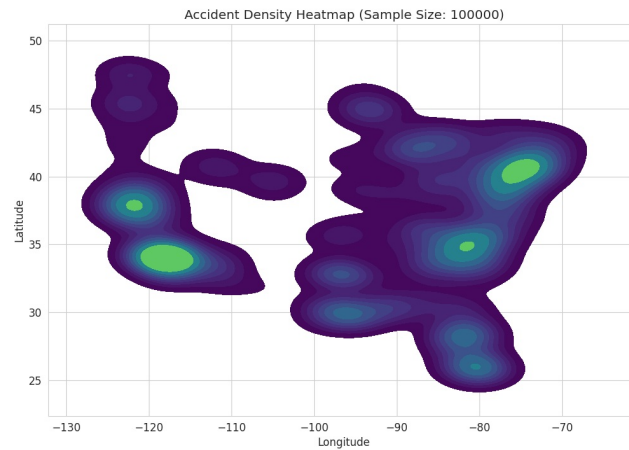
State Distribution



Accidents by State

- California has the highest number of accidents
- Texas and Florida follow closely

Accident Density

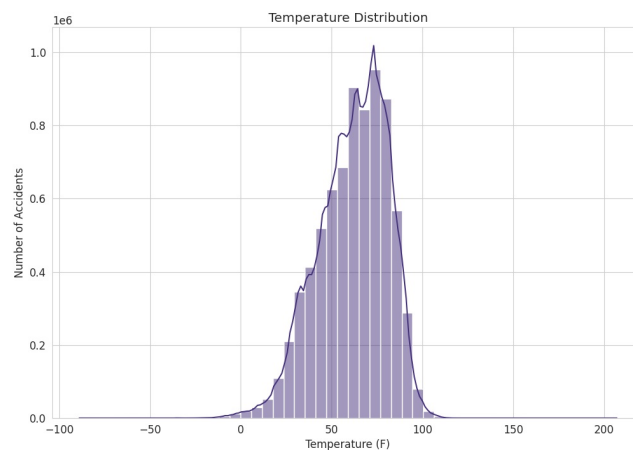


Accident Density

- Clear hotspots in major metropolitan areas
- Higher density along major highways
- Urban areas show significantly higher accident rates
- Rural areas show scattered, lower-density patterns

3. Weather Patterns

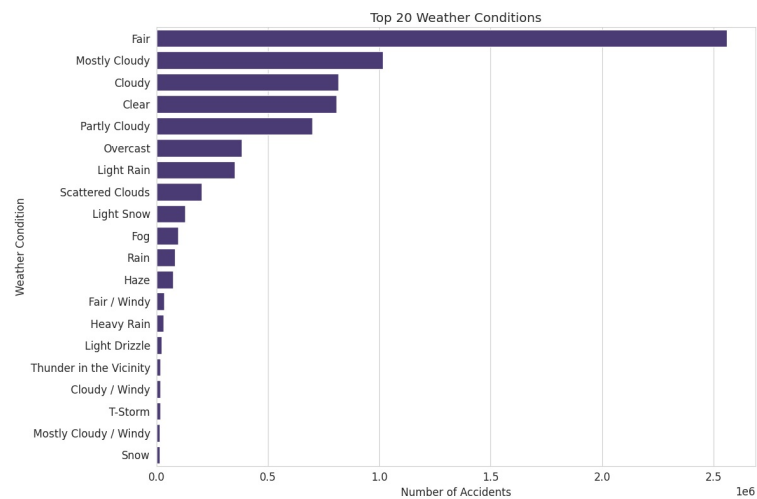
Temperature Impact



Temperature Distribution

- Most accidents occur in moderate temperatures (50-80°F)
- Fewer accidents in extreme temperatures
- Seems to be a normal distribution, temperature doesn't appear to have a huge impact

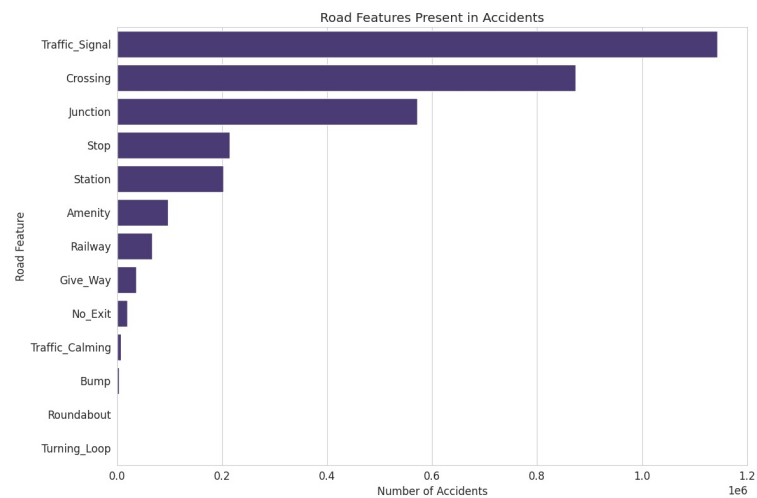
Weather Conditions



Weather Conditions

- Most accidents occur in clear weather
- Rain and cloudy conditions show high accident rates
- Snow and fog conditions show lower frequency but higher severity

4. Road Features

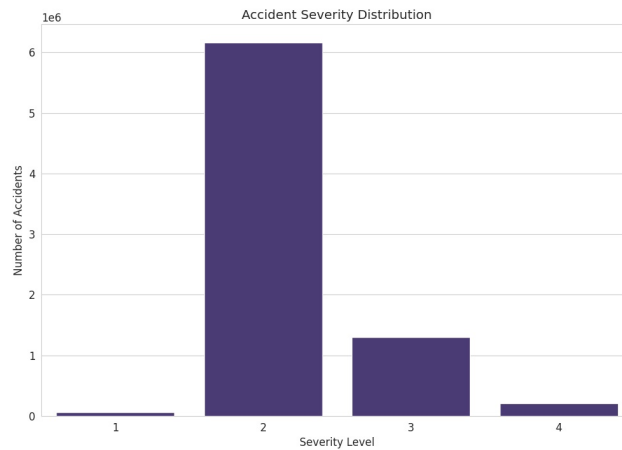


Road Features

- Traffic signals and junctions are most common accident locations
- Crossings and stop signs show high accident rates
- Roundabouts show lower accident rates than signals

5. Severity Analysis

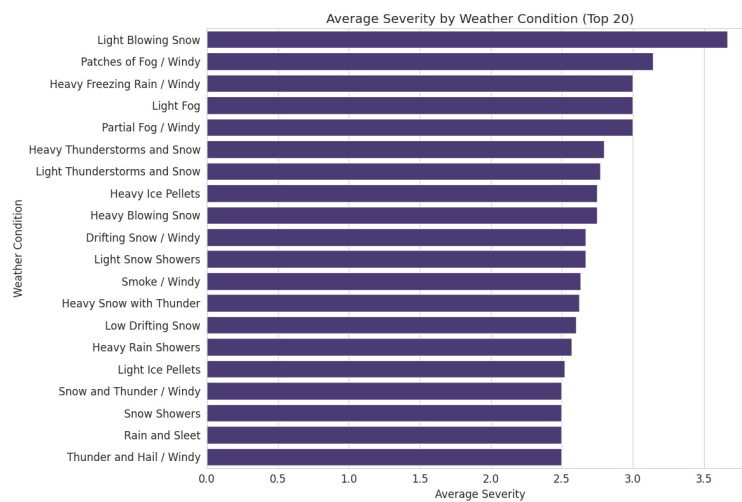
Severity Distribution



Severity Distribution

- Most accidents are of moderate severity (Level 2)

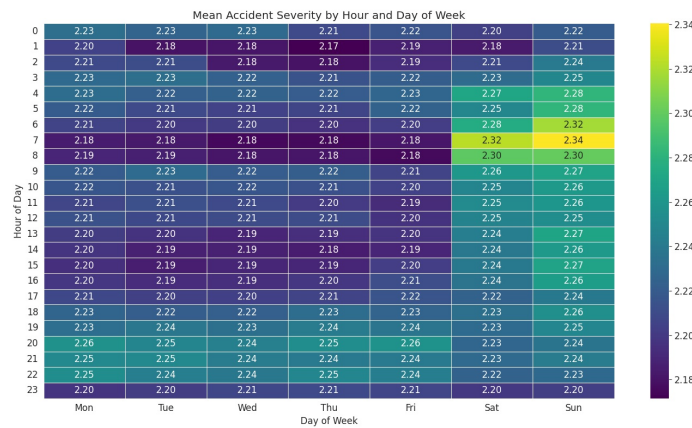
Severity by Weather



Severity by Weather

- Light blowing snow leads to most severe accidents
- All severe weather shown seems to cause more severe accidents

Mean Severity Heatmap (Hour vs. Day of Week)



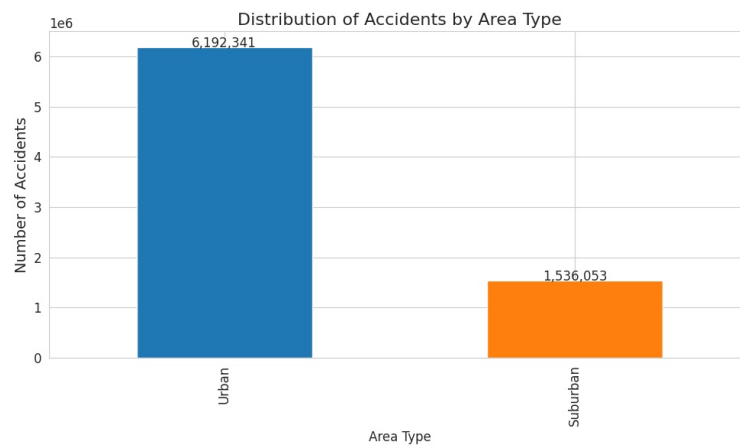
Mean Severity by Hour and Day of Week

- This heatmap shows the average severity level for accidents occurring at specific hours on specific days.
- Saturday and Sunday morning accidents tend to be most severe
- Overall, Sundays have the most severe accidents

6. Urban vs. Suburban Analysis

This section compares accident patterns between areas classified as Urban and Suburban using the GeoPandas spatial join.

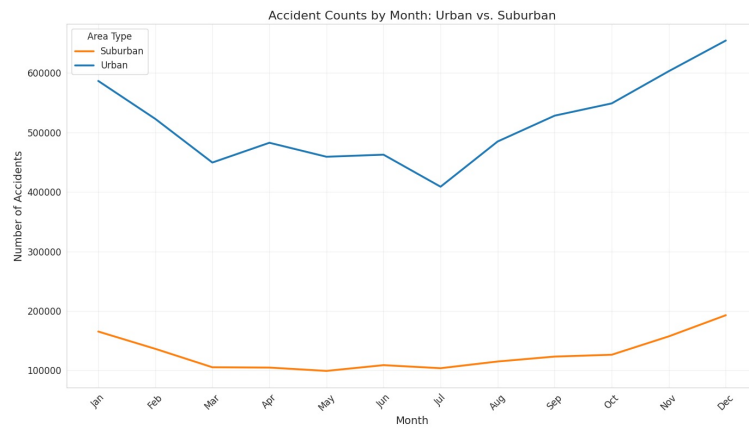
Area Type Distribution



Distribution by Area Type

- Most accidents occur in urban areas

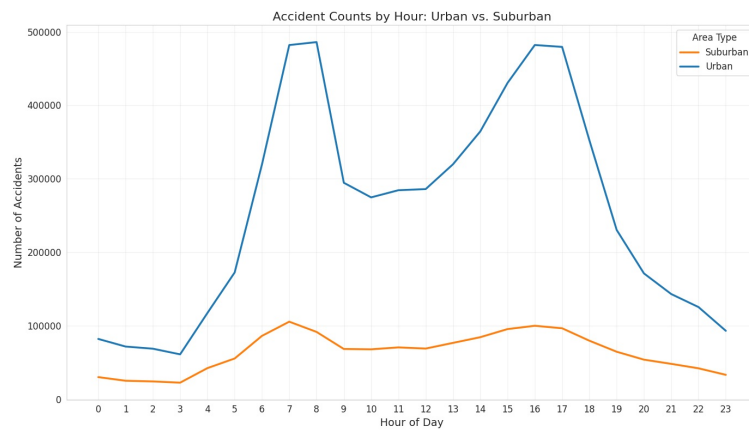
Timing Analysis (Monthly)



Accident Counts by Month

- By month, urban and suburban both follow a similar trend

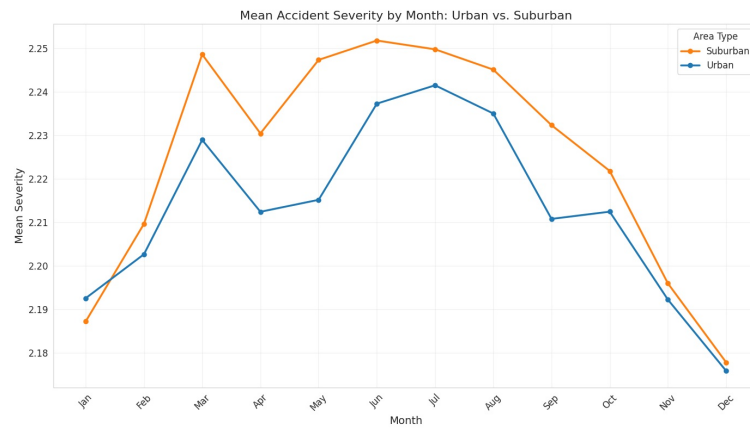
Timing Analysis (Hourly)



Accident Counts by Hour

- Both urban and suburban follow a similar trend, but urban areas are greatly exaggerated during commute hours as people are getting to and leaving work in the city

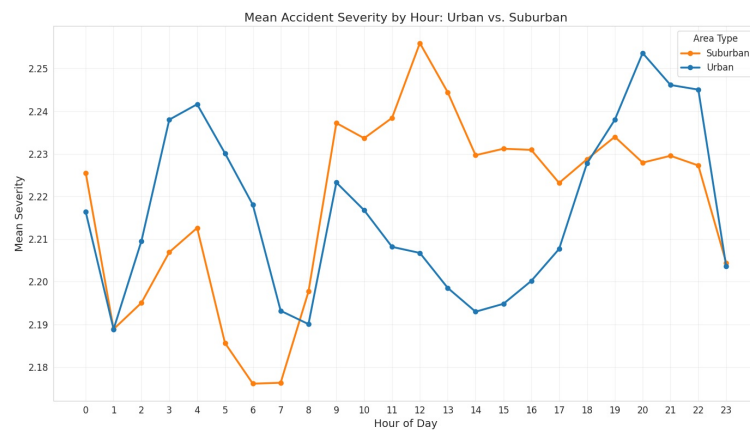
Severity Analysis (Monthly)



Mean Severity by Month

- Severity in urban and suburban are similar, with urban experiencing a noticeable dip in September

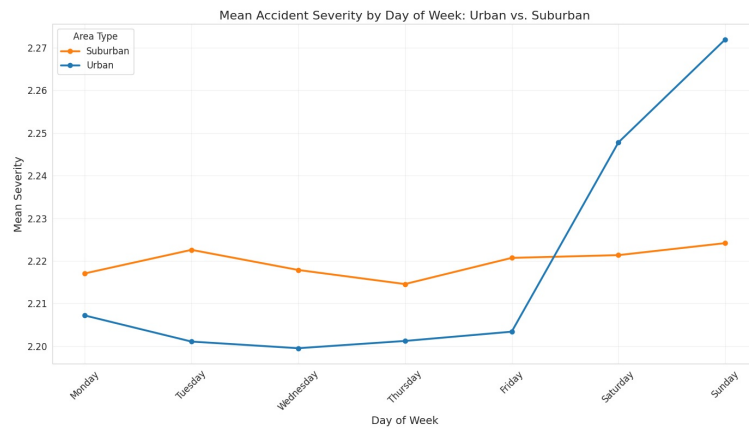
Severity Analysis (Hourly)



Mean Severity by Hour

- Urban areas typically have less severe accidents during the workday while suburban severity rates are higher over the same time
- Suburban areas have a spike in severity around noon (probably due to people driving to get lunch)

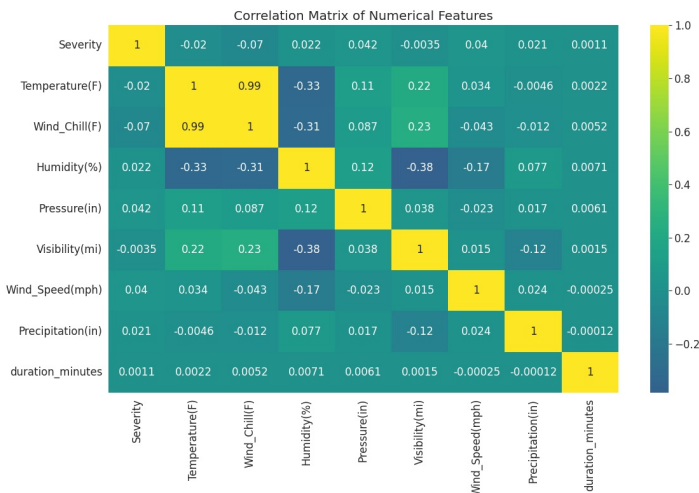
Severity Analysis (Day of Week)



Mean Severity by Day of Week

- Urban areas tend to have more severe accidents during weekends - likely due to activities, bars, restaurants, etc.

7. Correlation Analysis



Correlation Matrix

Key Correlations

- Strong positive correlation between:
 - Temperature and Wind Chill
 - Visibility and Precipitation
- Moderate negative correlation between:
 - Temperature and Humidity
 - Visibility and Wind Speed
- Weak correlation between:
 - Severity and most weather features
 - Duration and weather conditions

This doesn't seem to bring any real insights, as these weather features typically occur together.

8. Key Insights

1. **Temporal Factors:**
 - Rush hours and weekdays are high-risk periods
 - Winter months show increased accident rates
 - Early morning hours show lower accident rates
2. **Geographical Factors:**
 - Urban areas and major highways are high-risk locations
 - States with high population density show more accidents
 - Clear regional patterns in accident distribution
3. **Weather Impact:**
 - Most accidents occur in clear weather
 - Rain and cloudy conditions show high accident rates
 - Snow and fog conditions show lower frequency but higher severity
4. **Road Features:**
 - Traffic signals and junctions are critical locations
 - Road features designed for safety show lower accident rates
 - Infrastructure plays a significant role in accident patterns
5. **Severity Patterns:**
 - Weather conditions strongly influence accident severity
 - Time of day impacts severity levels
 - Most accidents are of moderate severity
6. **Urban vs. Suburban Differences:**
 - Urban areas account for the majority of accidents.
 - Urban accident counts peak significantly during commute hours compared to suburban areas.
 - Monthly trends for accident counts and severity are broadly similar between urban and suburban areas.
 - Urban accidents tend to be *less* severe during typical workday hours.
 - Urban accidents tend to be *more* severe on weekends.
 - Suburban areas experience higher average severity during the workday, with a notable peak around noon.